AUC DL Lab at AraSeg Shared Task 2026: Syntax-Aware Graphs, Bidirectionalized Decoders, and Encoders for Arabic Sentence Segmentation

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Abstract

The AUC DL Lab system addresses the AraSeg 2026 shared task on Arabic sentence segmentation across four conditions: paragraph-aware (PA), no-paragraph (NP), no-punctuation paragraph-aware (NOPNX-PA), and no-punctuation no-paragraph (NOPNX-NP), in both the closed and open tracks. We cast segmentation as imbalanced word-level boundary classification and pair fine-tuned Arabic encoders with two complementary architectures: HDP-GRAPH, which models syntactic structure via dependency-graph message passing, and a quantized decoder-only LLM using Just Pass Twice (JPT) with QLoRA for contextual boundary classification. Predictions are calibrated and combined through differential-evolution ensemble weighting. On the blind test set, our closed-track systems reach 94.5, 90.8, 89.2, and 86.0 F1 (ranking 5th, 6th, 6th, 7th), and our open-track systems reach 94.6, 90.9, 89.7, and 86.0 F1 (ranking 3rd, 4th, 4th, 5th) on PA, NP, NOPNX-PA, and NOPNX-NP, respectively. Architectural diversity grows more valuable as punctuation and paragraph cues are removed, while external data yields only marginal gains once in-domain coverage is adequate.

1 Introduction

Sentence segmentation, the partitioning of running text into sentence-like units, is a foundational preprocessing step for machine translation, parsing, summarization, and retrieval. Punctuation-based segmentation can be challenging for Arabic, particularly in unpunctuated text, where explicit sentence-boundary cues may be absent (Alshanqiti et al., 2022). In such cases, segmentation may require linguistic and contextual cues beyond punctuation.

The AraSeg 2026 shared task (Elkholy et al., 2026b) evaluates segmentation under four conditions with progressively fewer surface cues: paragraph-aware (PA), no-paragraph (NP), no-punctuation paragraph-aware (NOPNX-PA), and no-punctuation no-paragraph (NOPNX-NP), each with a *closed* track (official data only) and an *open* track (any public resource). The task is hardest under NOPNX-NP, where models must recover boundaries from lexical, syntactic, and discourse information alone, and is compounded by three properties: **class imbalance** (boundaries are rare, so detection is sensitive to the loss and threshold), **long documents** (they exceed encoder context windows, requiring windowed processing and cross-window aggregation), and **cue removal** (strong local signals vanish, forcing models to capture longer-range structure).

We make four contributions. First, we develop a *single cue-aware framework* that adapts boundary prediction to the surface cues available in each condition. Second, we introduce two complementary architectures: HDP-GRAPH, which augments an Arabic encoder with relation-weighted graph convolutions over Universal Dependencies parses, and a bidirectionalized decoder LLM adapted for per-token boundary tagging with 4-bit QLoRA. Third, we combine these with a calibrated differential-evolution ensemble and a leakage-guarded pipeline for external data in the open track. Fourth, our analysis shows that architectural diversity becomes increasingly valuable as surface cues are removed, while external data provides limited gains when in-domain coverage is strong. We release our hyperparameters, prompts, and Gemini document mapping here.

2 Background

2.1 Task Description

Each instance is an Arabic document tokenized into words $x_{1:n}$; the system predicts a binary sequence $y_{1:n}$ with $y_w \in \{0, 1\}$, where $y_w{=}1$ iff a sentence boundary follows word w. For

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example, for a five-word span $\langle w_1\, w_2\, w_3\, w_4\, w_5\rangle$ whose second and fifth words are sentence-final, the gold sequence is $\langle 0, 1, 0, 0, 1\rangle$. Systems are scored by boundary precision, recall, and F_1, macro-averaged over documents (Elkholy et al., 2026b,a). The four conditions share the same underlying documents and differ only by deterministic transformation: NP removes paragraph (newline) markers; NOPNX-PA removes punctuation; NOPNX-NP removes both. A model must therefore remain robust as surface cues are withdrawn. The *closed* track permits only the official training data; the *open* track permits any public resource. We submit to all four conditions in both tracks. All model and threshold selection is performed on the development set; the blind test set is used only for the final, one-shot submission that produces the official ranking.

2.2 Related Work

Sentence Segmentation. Classical systems rely on punctuation and hand rules. Neural segmentation models include Ersatz (Wicks and Post, 2021), which targets punctuated text, and punctuation-agnostic approaches such as WTP (Minixhofer et al., 2023) and SAT (Frohmann et al., 2024), which report strong multilingual results. Methods that rely on newline and punctuation statistics can degrade when both punctuation and paragraph structure are absent. For unpunctuated Arabic, PDTS predicts missing punctuation using a multilingual BERT-based model and validates the predictions with linguistic rules (Alshanqiti et al., 2022). The AraSeg corpus benchmarks encoder, dependency-parser, and LLM segmenters (Elkholy et al., 2026b).

Arabic PLMs. We build on AraBERT (Antoun et al., 2020), AraELECTRA (Antoun et al., 2021), and CAMeLBERT (Inoue et al., 2021), with a decoder LLM from the Qwen3 family (Yang et al., 2025).

Bidirectionalizing Decoders. Causal LLMs can be turned into strong encoders by removing the causal mask (BehnamGhader et al., 2024) or, without architectural change, by repeating the input so later copies attend over the whole sequence (Ewais et al., 2026). Prior work extracts *pooled* embeddings; we instead read *per-token* states for boundary tagging.

Structure-Aware Tagging. Graph convolutions over dependency trees (Li et al., 2022) propagate long-range syntactic signal that purely local taggers miss, motivating HDP-GRAPH.

3 System Overview

3.1 Shared Boundary-Tagging recipe

All encoder models tag words using the contextual representation of each word's first sub-word. Since documents exceed the encoder context, we process each in overlapping fixed-length windows and average the per-word boundary logits over all windows containing that word. Boundaries are rare (8.3% of positions, $\approx$ 11:1 imbalance; Appendix A.1), so we use label-smoothed focal loss (Lin et al., 2020) We scale the focal term by four multiplicative per-word weights and, in training only, a hard-example weight:

$$\mathcal{L}_w^{\text{final}} = h_{d,w}\; b_w\; s_w\; \beta_{\tau(w)}\; \mathcal{L}_w^{\text{FL}}, \tag{1}$$

$$\mathcal{L}_{\text{batch}} = \frac{\sum_{w\in\mathcal{V}} \mathcal{L}_w^{\text{final}}}{|\mathcal{V}|}, \tag{2}$$

where d indexes the document and $\mathcal{V}$ the valid (non-padding) positions. The weights are: a boundary-type weight $\beta_{\tau(w)}$ up-weighting word-based over punctuation-cued boundaries; a window-tail weight s_w boosting the last 20% of a window; a pre-newline weight b_w boosting positions before a retained newline; and a mistake-count weight $h_{d,w}$ up-weighting repeatedly misclassified positions. Full definitions and hyperparameters (τ_{pos}=1.3, B_{nl}=3.0, λ=0.5, M=5) are in Appendix B, Eqs. 6–7.

We add fast-gradient adversarial perturbations on the embedding layer (Miyato et al., 2017) with perturbation scale ϵ=1.0; clean and adversarial gradients are accumulated before each step. In punctuated conditions we randomly mask punctuation during training. A paragraph-forcing rule sets y_w=1 at retained newline markers, which are near-deterministic boundaries (§5.2). Per model we tune a decision threshold on dev and average checkpoints ("souping") (Wortsman et al., 2022). Remaining weight definitions, the discriminative layer-wise learning rate, and the R-Drop alternative (Liang et al., 2021; Howard and Ruder, 2018) are in Appendix B.

3.2 Base Models

Arabic Encoders. CAMeLBERT-MSA, AraBERTv2 (base/large), and AraELECTRA are

fine-tuned as token classifiers under the shared recipe.

HDP-GRAPH (syntax-aware graph). We parse each document with the Stanza Arabic UD pipeline (Qi et al., 2020) and build a token graph whose edges are dependency arcs plus sequential (⟨SEQ⟩) and self (⟨SELF⟩) links. The 36 raw relations are mapped to 11 coarse buckets that control weight sharing: eight dependency buckets (core args, oblique/adverbial, nominal modifiers, coordination, function words, compound/MWE, punctuation, and root/other), the sequential (⟨SEQ⟩) and self (⟨SELF⟩) edge types, and an unknown-relation bucket (⟨UNK⟩). On top of encoder features we apply relation-weighted graph convolutions:

$$\boldsymbol{h}_i^{(l+1)} = \sigma\Big(\boldsymbol{W}_s \boldsymbol{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \omega_{r(i,j)}\, \boldsymbol{W}_r \boldsymbol{h}_j^{(l)}\Big), \quad (3)$$

where $\boldsymbol{h}_i^{(l)}$ is token i's representation at layer l, σ a nonlinearity, $\mathcal{N}(i)$ the neighbors of i, $r(i,j)$ the relation bucket of edge (i,j), $\omega_{r(i,j)}$ its learned scalar weight, and $\boldsymbol{W}_s$, $\boldsymbol{W}_r$ the self- and neighbor-transformation matrices, with residual connections and layer normalization.

JPT (bidirectionalized LLM). We adapt repetition-based bidirectionalization (Ewais et al., 2026) to sequence labeling. A length-n input is duplicated as $[x; x]$; under causal attention the second copy attends over the first, making its per-token states bidirectional. A linear head predicts

$$\hat{y}_w = \sigma(\boldsymbol{w}^\top \boldsymbol{z}_{n+w} + b), \quad (4)$$

where $\boldsymbol{z}_{n+w}$ is the hidden state of the w-th token of the second copy from Qwen3-4B, $\boldsymbol{w}$ and b the head's weight and bias, σ the sigmoid, and $\hat{y}_w$ the predicted boundary probability. The model is fine-tuned with 4-bit QLoRA (Dettmers et al., 2023; Hu et al., 2022) and optional DoRA (Liu et al., 2024). We classify every second-copy token rather than pooling one vector.

3.3 Calibration and Ensembling

Per-model probabilities are calibrated by temperature scaling or isotonic regression (Guo et al., 2017), then combined as

$$\bar{p}_w = \sum_m \lambda_m\, \tilde{p}_w^{(m)}, \quad (5)$$

where $\tilde{p}_w^{(m)}$ is model m's calibrated probability, λ_m its ensemble weight ($\sum_m \lambda_m = 1$), and $\bar{p}_w$ the combined probability. Weights are searched by differential evolution on dev (Storn and Price, 1997), which selected near-uniform averaging. A global threshold τ gives $\hat{y}_w = \mathbb{1}[\bar{p}_w \geq \tau]$.

3.4 Cue-Aware Conditioning

The same framework is reconfigured per condition (Table 1). Paragraph forcing and boundary-type weighting apply only when their cue exists; once punctuation is stripped, every boundary is word-based and those components become inert.

Component	PA	NP	NOPNX-PA	NOPNX-NP
Paragraph forcing	✓	–	✓	–
Boundary-type weight	✓	✓	–	–
Punctuation masking	✓	✓	–	–
Focal loss + FGM	✓	✓	✓	✓
Calibration	–	–	✓	✓

Table 1: Cue-aware activation of recipe components by condition, used in both open and closed tracks.

3.5 Open-Track Data Pipeline

For the open track we draw data from CAMeL Treebank and BAREC. A smart-selection filter applies a boundary-density gate, then greedily selects documents to reduce the base corpus's vocabulary and boundary-token deficiencies, re-scoring after each pick (Appendix B, Eq. 9). Selection is capped by an equal per-source boundary-token budget to avoid diluting the positive class (external corpora run 2.5–5% density versus PA's 8.3%). Condition-specific transforms project the selected documents into each sub-task. A two-layer normalized-hash and fuzzy n-gram leakage guard removes any overlap with the dev, test, or blind test sets.

4 Experimental Setup

Data and Metric. We use the official AraSeg data (identical source documents across all four conditions), split into 174 training, 222 development, and 262 test documents, with the final ranking computed on a separate blind test set. All conditions are scored with per-document boundary precision, recall, and macro-averaged F_1 (Elkholy et al., 2026b). Every ensemble and decision threshold is selected on development F_1; the blind test set is never inspected during development.

Preprocessing. Documents are used as released without re-tokenization. Encoders read them in

overlapping fixed-length windows and average per-word logits across windows (§3.1); open-track external corpora are cleaned to PA's character inventory and re-projected per condition (§3.5).

Models and libraries. Encoders (CAMeLBERT-MSA, AraBERTv2 base/large, AraELECTRA) are fine-tuned with HuggingFace Transformers; the decoder LLM uses PEFT and bitsandbytes for 4-bit QLoRA; dependency parses come from Stanza's Arabic UD pipeline (Qi et al., 2020); calibration uses scikit-learn; and the differential-evolution ensemble search uses SciPy (Storn and Price, 1997). Training-objective and data-selection formulas are in Appendix B.

5 Results

Sub-Task	Closed Track (F_1)	Open Track (F_1)
PA	**94.5**	**94.6**
NP	90.8	90.9
NoPnx-PA	89.2	89.7
NoPnx-NP	86.0	86.0

Table 2: Official blind test set F_1 scores across tracks and sub-tasks.

The scores follow the trend shown in the (Elkholy et al., 2026a) paper with model performance decreasing with each sub-task's removal of signal carrying tokens.

5.1 Key Empirical Findings

Component / Setup	Average Gain (ΔF_1)
Component Validation (PA Closed Baseline)	
HDP-Graph vs. Backbone Base	**+3.47**
Structural Forcing Rule	**+2.296**
Cross-Track System Effects (Test Set)	
Ensemble vs. Best Single Model	**+0.9425**
External Data Integration	**+0.33**

Table 3: Average test set gains (ΔF_1) for core system components (validated on the PA Closed baseline and deployed across all tracks) alongside cross-track ensemble and external data evaluations.

As shown in table 3, during the initial testing, enforcing sentence boundaries at paragraph delimiters (\n) through structural rule forcing consistently boosted baseline performance across models (Appendix C.4). Additionally, augmenting an encoder backbone with an HDP dependency graph component yielded substantial gains over the baseline (Appendix C.2). Combining several model architectures through ensembling consistently improved performance across all sub-tasks (Appendix C.1).Incorporating external data in the Open track provided modest improvements relative to the Closed track setup (Appendix C.3).

5.2 Error Analysis

Upon examination of the confidence distributions across the different tracks and sub-tasks, correct predictions exhibit a strongly U-shaped confidence distribution, concentrating near probabilities of 0 and 1, indicating a decisive classifier (Appendix D.1). Incorrect predictions are disproportionately associated with punctuation tokens and concentrate near decision thresholds, reflecting greater ambiguity (Appendix D.2). Complete signal removal truncates the upper confidence range, reducing the model's ability to make high-confidence decisions (Appendix D.3). Exploratory genre-level analysis suggests that the Religion and Wiki categories show the largest performance drops as cues are removed (Appendix C.5).

6 Conclusion

We presented the AUC DL Lab submission to the AraSeg 2026 shared task, framing Arabic sentence segmentation as imbalanced word-level boundary tagging across all four conditions and both tracks. Our single cue-aware framework pairs standard Arabic encoders with two complementary architectures, a syntax-aware dependency-graph model (HDP-GRAPH) and a bidirectionalized decoder LLM (JPT + QLoRA), through a calibrated differential-evolution ensembling. Three findings recur across conditions: architectural diversity becomes increasingly valuable as punctuation and paragraph cues disappear, with ensemble gains widening under weaker surface cues; forcing boundaries at retained paragraph markers provides a near-free, high-yield correction; carefully selected, leakage-guarded external data adds little beyond in-domain supervision. Confidence and error analyses further show that punctuation disproportionately supports performance in the hardest genres. The cue-aware recipe and complementary architectures provide a starting point for punctuation-robust segmentation in Arabic and other low-punctuation scripts.

Limitations

The main limitations of this work come from the computational overhead and the constrained experimental coverage. Ensemble construction incurs significant training costs compared to a single model. Several weighting, threshold, and ensemble hyperparameters were selected through development-set experimentation rather than systematic hyperparameter optimization. Furthermore, the effects of the HDP-Graph were only evaluated on one backbone, and our training pipeline was developed and tuned primarily on the PA Closed track and then applied unchanged to the remaining tracks, so it was not separately optimized for each condition. Finally, the impact of external data was measured only on the BAREC and CAMeL Treebank datasets.

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A Data Statistics and Ensemble Composition

A.1 Global Sentence-Boundary Rate

Statistic	Value
Documents (train split)	174
Word tokens	128,173
Sentence boundaries	10,657
Non-boundary tokens	117,516
Boundary rate	8.3145%
Non-boundary : boundary	11.03 : 1

Table 4: Training-set statistics used to derive the global sentence-boundary rate and class ratio reported in Section 3.1. Computed exclusively over the training split.

A.2 Ensemble Composition by Sub-Task & Track

Sub-Task & Track	Model	Dev F1	Test F1	Blind Test F1
PA Closed	CAMeLBERT-MSA	93.15%	92.74%	x
	AraBERTv2 (base)	91.52%	90.70%	x
	AraBERTv2-large	91.45%	90.76%	x
	AraELECTRA	93.99%	93.30%	x
	Qwen3-4B (JPT + QLoRA)	92.77%	92.30%	x
	HDP-Graph	94.09%	93.39%	x
	Ensemble (DE, equal-weight)	**94.78%**	**94.04%**	**94.5%**
NP Closed	CAMeLBERT-MSA	91.09%	91.06%	x
	AraBERTv2-large	90.00%	89.21%	x
	AraELECTRA	92.02%	91.68%	x
	Qwen3-4B (JPT + QLoRA)	89.32%	86.48%	x
	HDP-Graph	92.14%	91.24%	x
	Ensemble (DE, equal-weight)	**93.21%**	**92.80%**	**90.8%**
NoPnx-PA Closed	CAMeLBERT-MSA	82.81%	83.96%	x
	AraBERTv2-large	79.74%	79.32%	x
	AraELECTRA	85.34%	85.40%	x
	Qwen3-4B (JPT + QLoRA)	78.34%	77.41%	x
	HDP-Graph	81.35%	81.49%	x
	Ensemble (DE, equal-weight)	**87.05%**	**86.71%**	**89.2%**
NoPnx-NP Closed	CAMeLBERT-MSA	80.75%	82.09%	x
	AraBERTv2-large	77.10%	75.99%	x
	AraELECTRA	83.74%	83.98%	x
	Qwen3-4B (JPT + QLoRA)	72.20%	72.02%	x
	HDP-Graph	80.45%	81.01%	x
	Ensemble (Isotonic Cal. + DE, equal-weight)	**85.01%**	**84.78%**	**86.0%**
PA Open	CAMeLBERT-MSA (Open)	93.69%	93.52%	x
	AraELECTRA (Open)	94.87%	94.63%	x
	AraELECTRA (Closed)	93.99%	93.30%	x
	HDP-Graph (Closed)	94.09%	93.39%	x
	Ensemble (Isotonic Cal. + DE, equal-weight)	**95.21%**	**95.02%**	**94.6%**
NP Open	CAMeLBERT-MSA (Open)	91.60%	91.63%	x
	CAMeLBERT-MSA (Closed)	91.09%	91.06%	x
	AraELECTRA (Open)	92.34%	91.98%	x
	AraELECTRA (Closed)	92.02%	91.68%	x
	HDP-Graph (Closed)	92.14%	91.24%	x
	Ensemble (DE, equal-weight)	**93.46%**	**93.12%**	**90.9%**
NoPnx-PA Open	CAMeLBERT-MSA (Open)	83.24%	84.44%	x
	CAMeLBERT-MSA (Closed)	82.81%	83.96%	x
	AraELECTRA (Open)	85.38%	84.78%	x
	AraELECTRA (Closed)	85.34%	85.40%	x
	Ensemble (Temp.-Scaled DE, equal-weight)	**86.38%**	**86.26%**	**89.7%**
NoPnx-NP Open	Qwen3-4B (Closed)	72.20%	72.02%	x
	Qwen3-4B (Open)	72.59%	72.24%	x
	AraBERTv2-large (Closed)	77.10%	75.99%	x
	AraBERTv2-large (Open)	79.24%	78.91%	x
	HDP-Graph (Closed)	80.45%	81.01%	x
	CAMeLBERT-MSA (Closed)	80.75%	82.09%	x
	CAMeLBERT-MSA (Open)	81.10%	82.47%	x
	AraELECTRA (Open)	83.54%	83.93%	x
	AraELECTRA (Closed)	83.74%	83.98%	x
	Ensemble (Isotonic Cal. + Temp.-Scaled DE, equal-weight)	**85.71%**	**85.25%**	**86.0%**

Table 5: Ensemble composition across all tracks, with individual and ensemble dev, test, and blind-test macro-F1.

B Training Objective and Data-Selection Formulas

Full definitions of the components summarized in §3; the combined per-token loss is Eq. 1.

B.1 Loss Formulation

Per-Word Weights. The boundary-type weight (priority comma > full-stop > other-punct) up-weights word-based over punctuation-cued boundaries:

$$\beta_{\tau(w)} = 1 + \sum_{k \in \mathcal{K}} (w_k - 1)\, \mathbf{1}[\text{bnd}_{w,k}], \tag{6}$$

with $\mathcal{K} = \{\text{word}, \text{comma}, \text{fs}, \text{punct}\}$, w_{word}=4.0, w_{comma}=4.5, w_{fs}=2.75, w_{punct}=3.0. The remaining factors boost the window tail (τ_{pos}=1.3), pre-newline positions (B_{nl}=3.0), and repeatedly misclassified positions (λ=0.5, cap M=5, mistake counter $m_{d,w}$). The five factors combine multiplicatively:

$$\begin{gathered} s_w = 1 + (\tau_{\text{pos}} - 1)\, \mathbf{1}[w \in \text{last } 20\%], \qquad b_w = 1 + (B_{\text{nl}} - 1)\, \mathbf{1}[\text{next-is-newline}], \\ h_{d,w} = 1 + \lambda \min(m_{d,w}, M). \end{gathered} \tag{7}$$

B.2 Smart Selection of External Data (Open Track)

Density Gate and Budget. Each candidate d must pass $\rho(d) = \frac{1}{|d|} \sum_j \mathbf{1}[l_j{=}1] \geq \rho_{\min}$. The added boundary budget $B_{\text{total}} = \lfloor r\, |\mathcal{B}_{\text{base}}| \rfloor$ (r=0.20) is split equally across the S sources, and all sources are scored against a fixed PA-only baseline (making selection order-independent): $B_s = \left\lfloor \frac{B_{\text{total}}}{S} \right\rfloor$ $(s < S)$, $\quad B_S = B_{\text{total}} - (S-1) \left\lfloor \frac{B_{\text{total}}}{S} \right\rfloor$ (8)

Greedy Deficiency Scoring (default). With running counts C_t, vocabulary target $T_t^{\text{vocab}} = C_t^{\text{base}}$, and boundary target $T_t^{\text{bnd}} = C_t^{\text{base,bnd}} + \Delta \cdot C_t^{\text{base,bnd}} / \sum_s C_s^{\text{base,bnd}}$ ($\Delta = \min(C_{\text{non-bnd}}^{\text{base}} - C_{\text{bnd}}^{\text{base}}, B)$; novel types default to T^{vocab}=1, T^{bnd}=5), each candidate scores

$$\text{score}(d) = \sum_{t \in \mathcal{V}(d)} x_t\, \alpha_{\text{vocab}} \max\Big(0, \tfrac{T_t^{\text{vocab}} - C_t}{T_t^{\text{vocab}}}\Big) + \sum_{t \in \mathcal{B}(d)} x_t\, \alpha_{\text{bnd}} \max\Big(0, \tfrac{T_t^{\text{bnd}} - C_t^{\text{bnd}}}{T_t^{\text{bnd}}}\Big), \tag{9}$$

with α_{vocab}=1.0, α_{bnd}=3.0. The highest-scoring document is selected and counts updated until the budget is exhausted or $\max_d \text{score}(d) \leq 0$. A

B.3 Hard-Document Upsampling and Mining

Upsampling. Hard documents are duplicated (k=2, so each appears 3×): $\mathcal{D}'_{\text{train}} = \mathcal{D}_{\text{train}} \cup k \times \mathcal{H}$. The hard set $\mathcal{H}$ is chosen by F1 rank (bottom p=25%), a combined z-scored difficulty over punctuation density, sentence-length variance, and boundary count, or a punctuation-density threshold (τ=0.5).

Epoch-Level Mining. Every 2 epochs, per-document difficulty $d_i = 1 - F1_i$ is converted to a sampling distribution by a temperature-scaled softmax (T=1.5): $w_i = d_i^T / \sum_j d_j^T$.

C Component Analysis and Ablations

C.1 Ensemble Gains

Sub-Task	Closed Track (ΔF_1)	Open Track (ΔF_1)
PA	**+0.65**	**+0.39**
NP	**+1.12**	**+1.14**
NoPnx-PA	**+1.31**	**+0.86**
NoPnx-NP	**+0.8**	**+1.27**
Average	**+0.970**	**+0.915**

Table 6: Ensemble performance gains test (ΔF_1) across sub-tasks for Closed and Open tracks by comparing the ensemble to the highest performing model on the test set.

C.2 Dependency Graph Potential

Model / Architecture	Dev (F_1)	Test (F_1)
aubmindlab/bert-base-arabertv02	90.22	89.92
HDP-Graph	**94.09**	**93.39**

Table 7: F_1 comparison between the AraBERT backbone alone and the same backbone augmented with the HDP dependency graph on the PA Closed track

C.3 Limited Benefit from External Data

Sub-Task	Dev (ΔF_1)	Test (ΔF_1)	Blind Test (ΔF_1)
PA	**+0.43**	**+0.98**	**+0.10**
NP	**+0.25**	**+0.32**	**+0.10**
NoPnx-PA	**–0.67**	**–0.45**	**+0.50**
NoPnx-NP	**+0.70**	**+0.47**	**0.00**

Table 8: Open track gain (ΔF_1) over Closed track on Dev, Test, and Blind Test sets across sub-tasks.

C.4 Structural Rule Forcing Gains

Model	w/o Force (F_1)	w/ Force (F_1)	Gain (ΔF_1)
`AraBERTV2Base`	87.88	91.60	**+3.72**
`AraBERTV2Large`	90.37	92.75	**+2.38**
`AraBERTV02Base`	89.96	92.19	**+2.23**
`CAMeLBERT-MSA`	90.42	92.22	**+1.8**
`AraELECTRA`	91.55	92.90	**+1.35**

Table 9: Impact of structural rule forcing during early exploratory testing on the PA Closed test set across selected baselines.

C.5 Exploratory Genre-Level Patterns

Genre	# Docs	PA (F_1)	NoPnx-NP (F_1)	ΔF_1 (Signal Loss)
Religion	74	0.974	0.790	-0.184
Wiki	2	0.961	0.860	-0.101
Politics	2	0.932	0.881	–0.051
Literature	48	0.876	0.828	–0.048
Education	67	0.956	0.910	–0.046
Media	28	0.919	0.888	–0.031
Children	41	0.884	0.858	–0.026
Macro Avg.	**–**	**0.929**	**0.859**	**–0.070**

Table 10: Impact of complete signal removal across genres: Macro F_1 comparison between Full Signal (**PA**) and No Punctuation / Paragraph Delimiter Removal (**NoPnx-NP**) on the test set across the Open and Closed tracks.

Genre labels were reconstructed via zero-shot Gemini 3.6 Flash prompting following the categories in Elkholy et al. (2026a).

D Error Analysis

Each figure shows predicted sentence-boundary probability by token count, separated into words and punctuation on the test set. The dashed line marks the model's decision threshold: tokens to the right are predicted as boundaries, and those to the left as non-boundaries.

D.1 U-Shaped Confidence Distribution

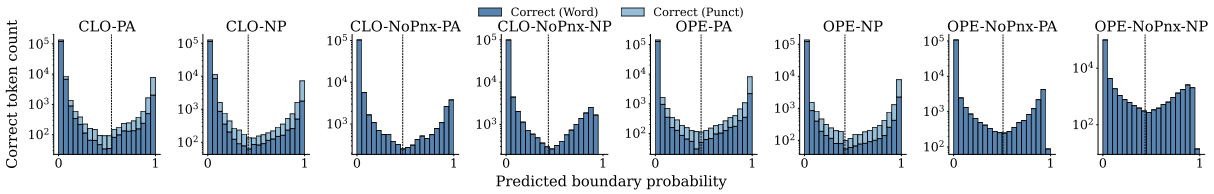


Figure 1: Confidence distribution of *correct* predictions, split into word tokens and punctuation tokens, across closed and open tracks for all four sub-tasks. Token counts are shown on a log scale due to class imbalance.

D.2 Punctuation Token Error Analysis

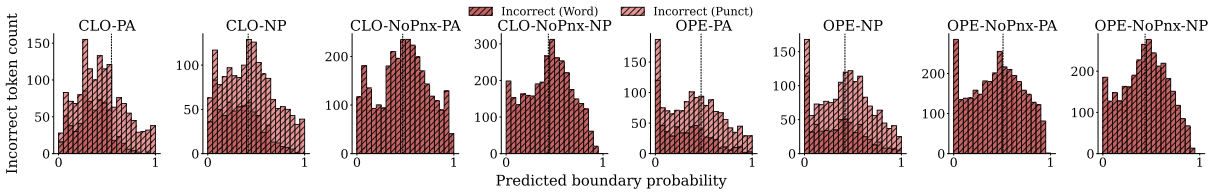


Figure 2: Confidence distribution of *incorrect* predictions, split into word tokens and punctuation tokens, across closed and open tracks for all four signal conditions. Punctuation-token errors are hatched to distinguish them from word-token errors. Token counts are shown on a linear scale.

D.3 Probability Truncation Effects

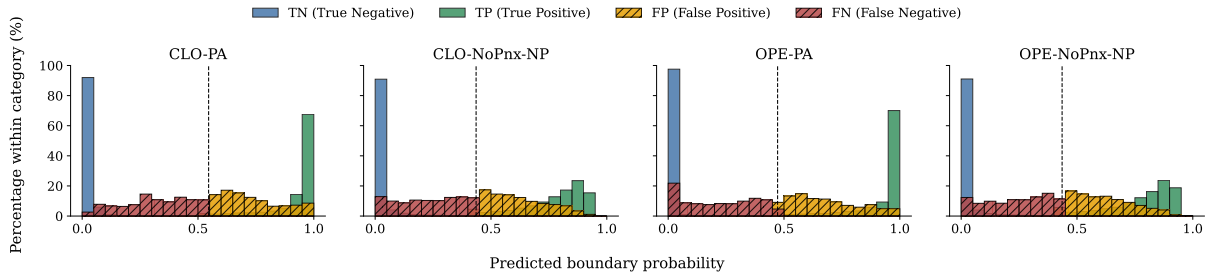


Figure 3: Per-category confidence distributions (TN, TP, FP, FN, each normalized to 100% within its own category) for the full-signal (**PA**) and fully-stripped (**NoPnx-NP**) conditions, across closed and open tracks. Contrasting these two endpoint conditions highlights the cumulative effect of removing both punctuation and paragraph markers.